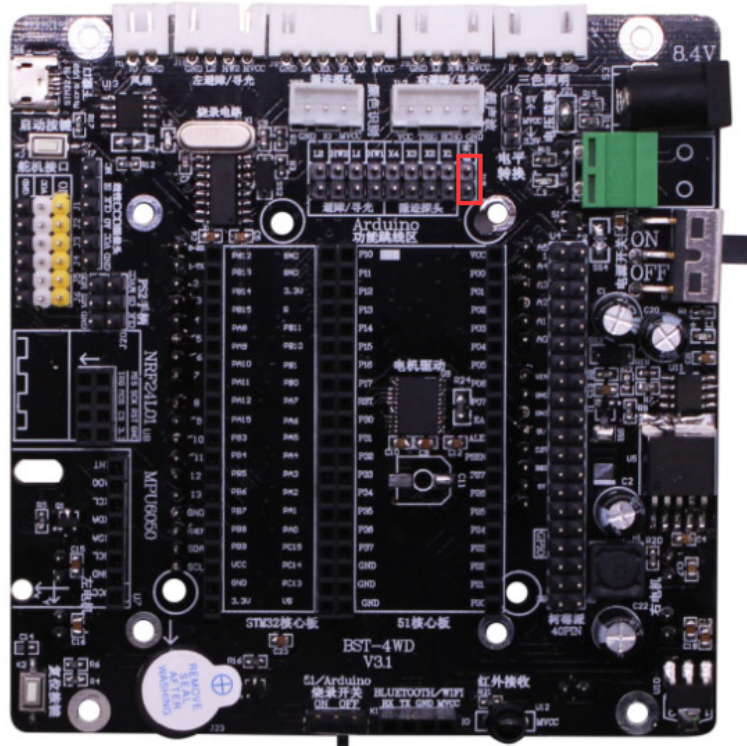


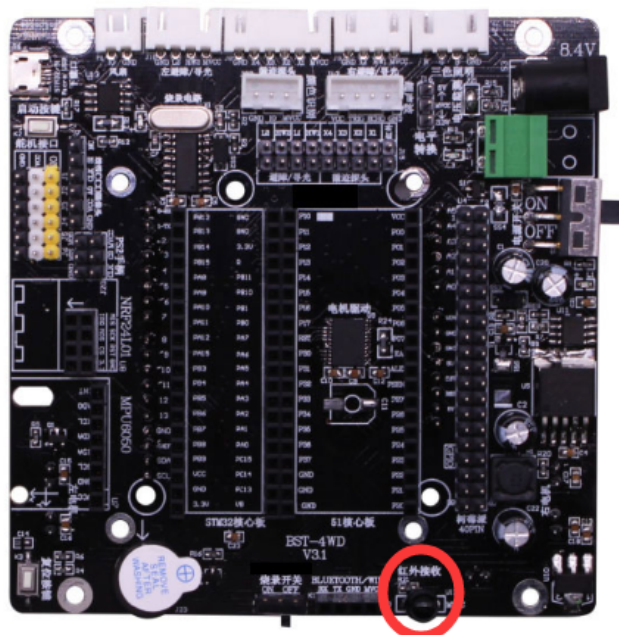
14. Arduino UNO platform ---- IR_control

!! Note:

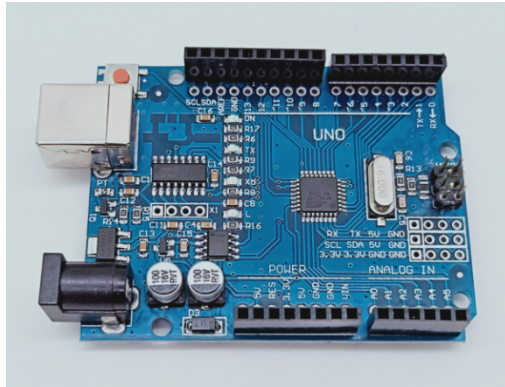
1. In order to avoid the interference of sunlight on the infrared sensor, this experiment needs to be carried out indoors.
2. In the infrared remote control experiment, there is a row of pins above the 51 core board on the expansion board, and the right two pin headers should be equipped with a jumper cap, as shown in the figure below.



3. We need to unplug the small fan and grayscale modules.
4. During remote control, the infrared remote control needs to face the infrared receiver on the expansion board.



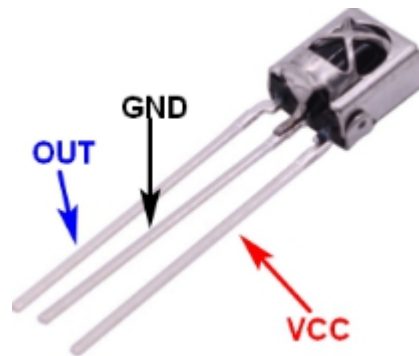
1) Preparation



1-1 Arduino UNO board



1-2 infrared remote control



1-3 infrared receiver

2) Purpose of Experimental

In this experiment, we used infrared remote communication to control the movement of the car.

3) Principle of experimental

User code: 0XFF



The infrared remote control receiving data is 00FFCMD (CMD inverted code). All key corresponding coding value as shown below:

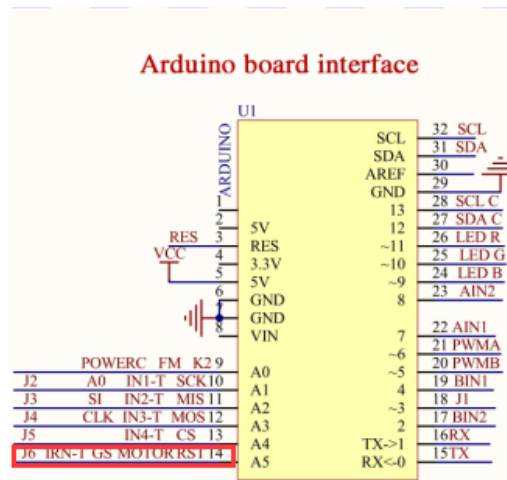
00FF00FF	
00FF30CF	

00FF708F	
00FF40BF	
00FFA05F	
00FF807F	
00FF20DF	
00FF609F	
00FF906F	
00FF10EF	
00FF50AF	
00FFB04F	
00FF08F7	
00FF5877	
00FF48B7	
00FF28D7	
00FFA857	
00FF6897	
00FF18E7	
00FF9867	
00FF58A7	

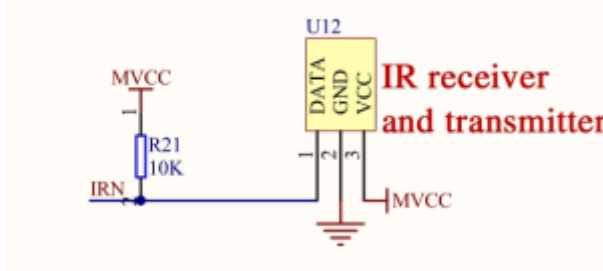
The function defined by each button can be customized by modifying the program.

4)Experimental Steps

4-1 About the schematic



4-1 Arduino UNO interface circuit diagram



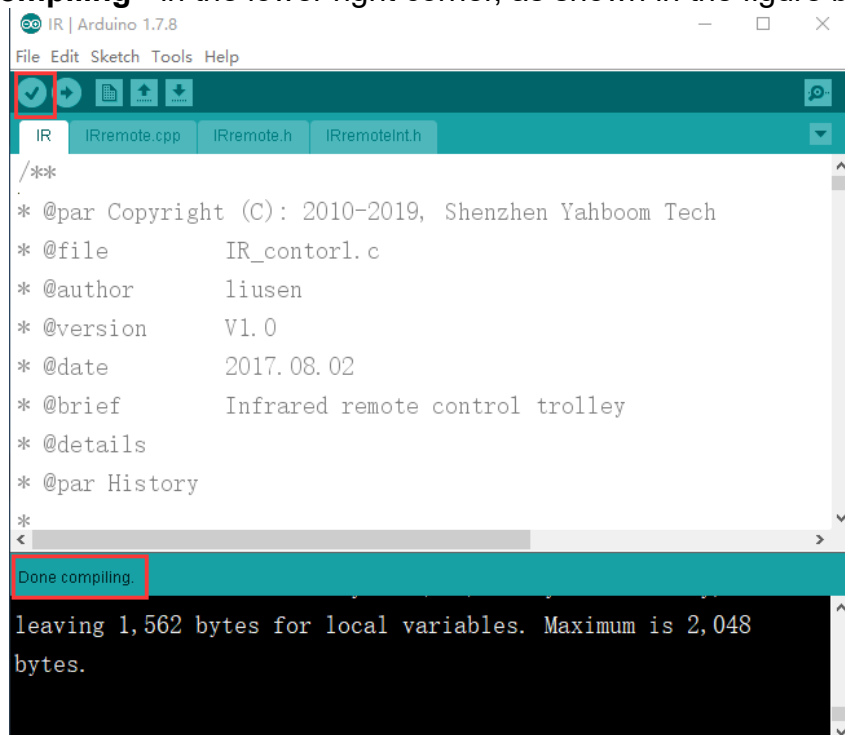
4-2 infrared receiver and transmitter

4-2 According to the circuit schematic:

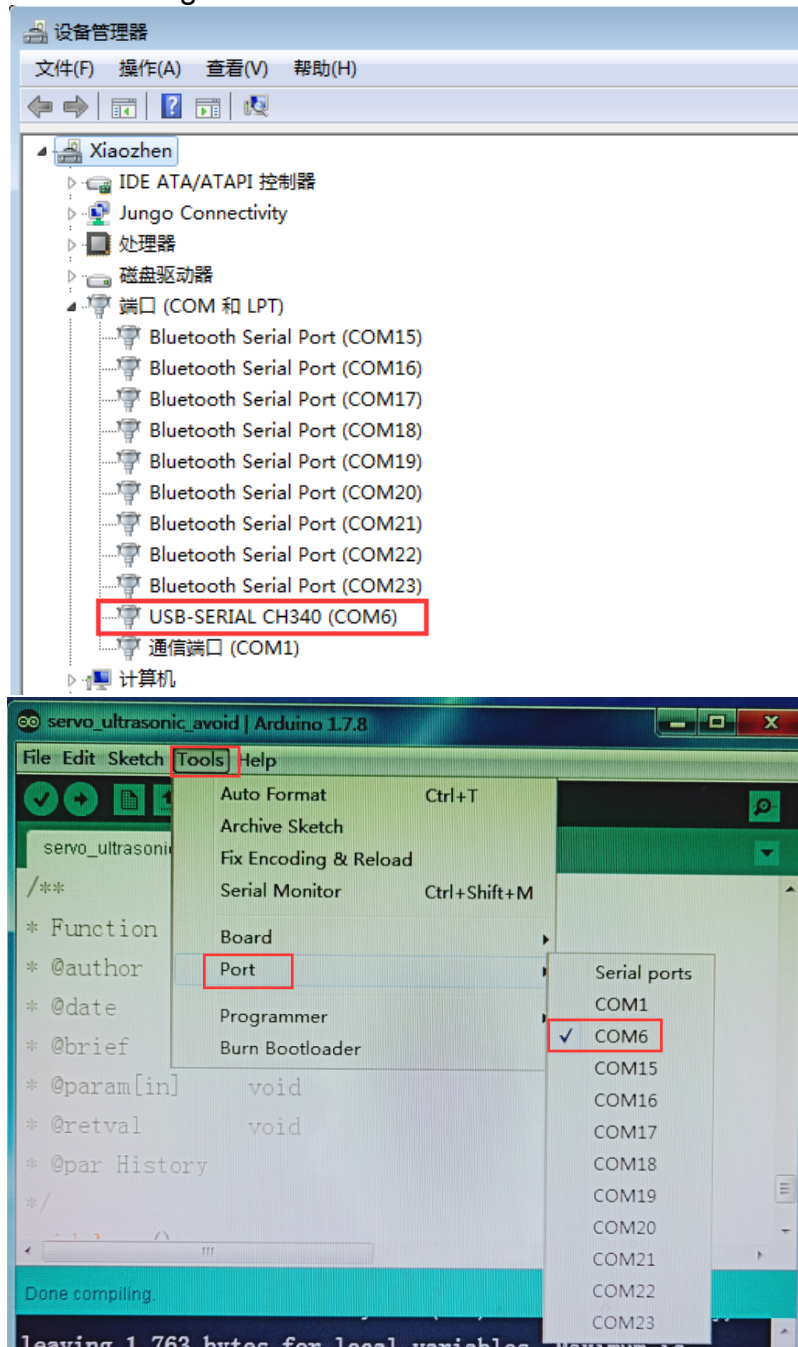
IRN-----A5(Arduino UNO)

4-3 About the code

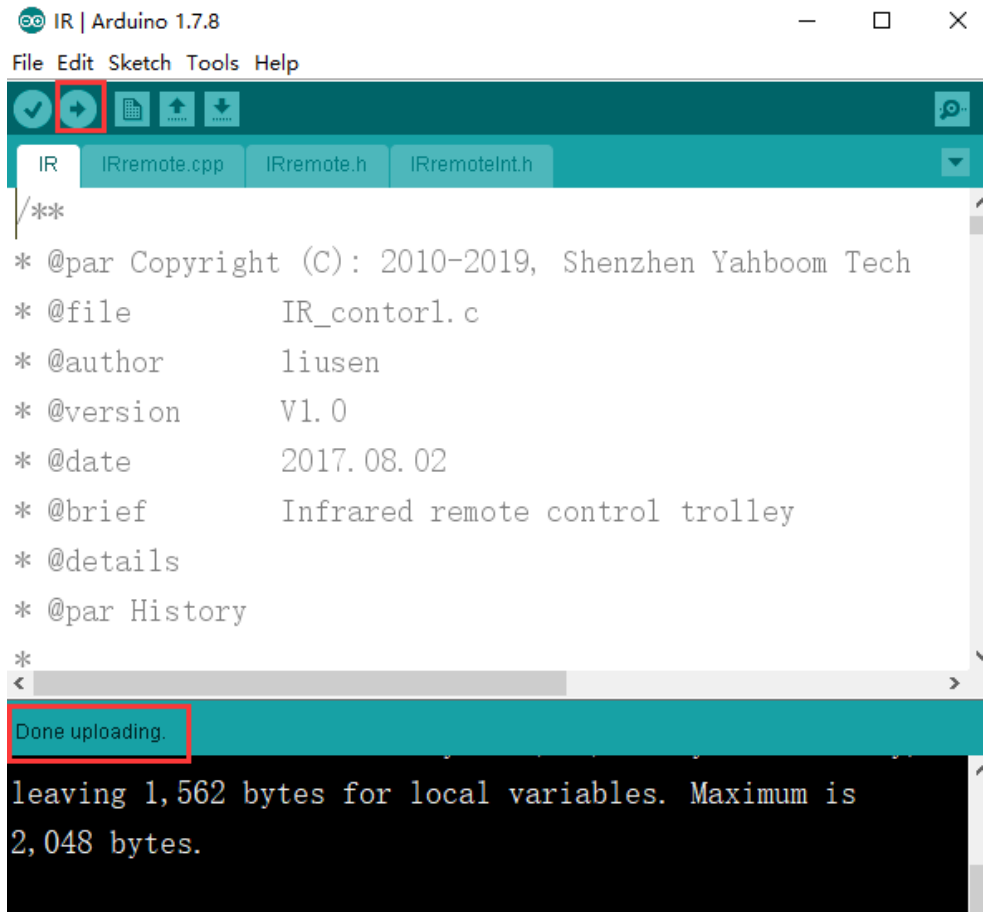
1. We need to open the code of this experiment: **IR_player_mode.ino** or **IR.ino**, click "✓" under the menu bar to compile the code, and wait for the word "Done compiling" in the lower right corner, as shown in the figure below.



2. In the menu bar of Arduino IDE, we need to select **Tools** --- **Port** --- selecting the port that the serial number displayed by the device manager just now, as shown in the figure below.



3. After the selection is completed, you need to click “→” under the menu bar to upload the code to the Arduino UNO board. When the word **“Done uploading”** appears in the lower left corner, the code has been successfully uploaded to the Arduino UNO board, as shown in the figure below.



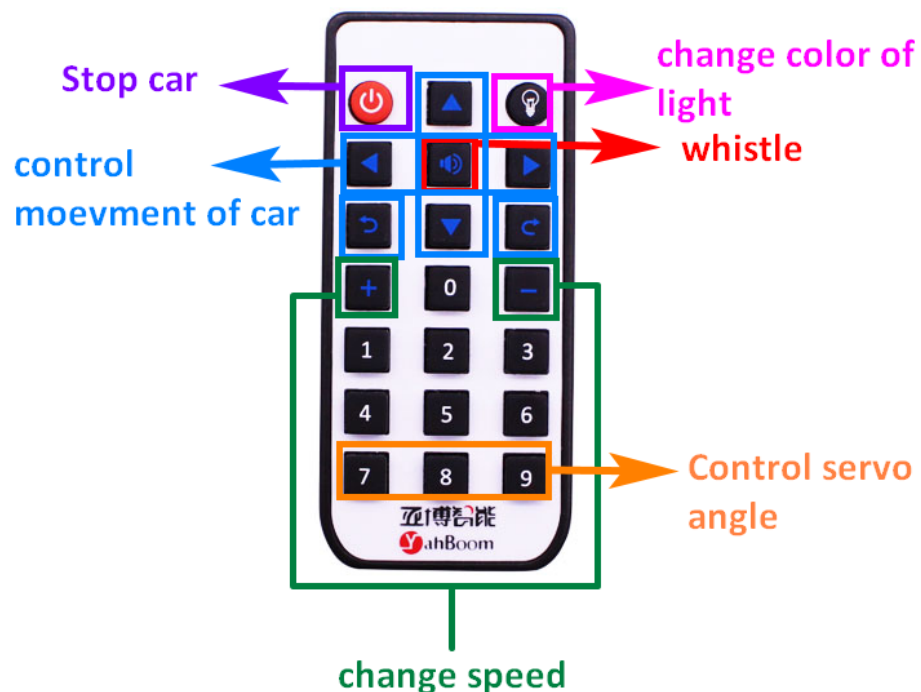
```

IR | Arduino 1.7.8
File Edit Sketch Tools Help
IR IRremote.cpp IRremote.h IRremoteInt.h
/*
 * @par Copyright (C): 2010-2019, Shenzhen Yahboom Tech
 * @file      IR_contorl.c
 * @author    liusen
 * @version   V1.0
 * @date      2017.08.02
 * @brief     Infrared remote control trolley
 * @details
 * @par History
 *
 */
Done uploading.
leaving 1,562 bytes for local variables. Maximum is
2,048 bytes.

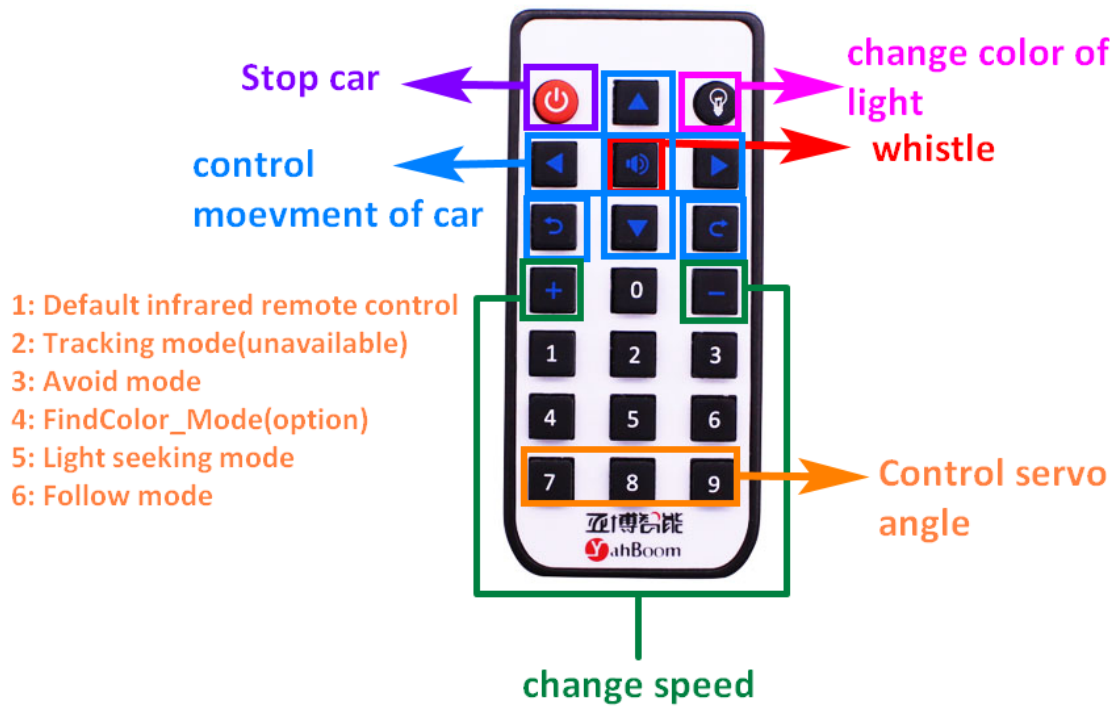
```

The following functions are controlled by the infrared remote controller:

About IR.ino code:



About IR_player_mode.ino:



!Note: Find color mode need to color track, please purchase additional.